

PVDeg: a Python package for modeling degradation of solar photovoltaic systems

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 05 December 2025

Published: unpublished

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Summary

PVDeg is an open-source Python package for modeling degradation of photovoltaic (PV) modules. It provides modular functions, materials databases, and calculation workflows for simulating degradation mechanisms (e.g., LeTID, hydrolysis, UV exposure) using weather data from the National Solar Radiation Database (NSRDB) and the Photovoltaic Geographical Information System (PVGIS). By integrating Monte Carlo uncertainty propagation and geospatial processing, PVDeg enables field-relevant predictions and uncertainty quantification of module reliability and lifetime.

PVDeg is developed openly on GitHub and releases are distributed via the Python Package Index (PyPI). The source code is freely available under the BSD 3-Clause license, and copyrighted by the Alliance for Sustainable Energy allowing permissive use with attribution. PVDeg follows best practices for open-source python software, with a robust testing framework across Python 3.x environments, semantic versioning, and supporting documentation available at pvdegradationtools.readthedocs.io. The package is developed at the National Laboratory of the Rockies (NLR), previously known as the National Renewable Energy Laboratory (NREL), and supported by the Durable Module Materials (DuraMAT) consortium.

As an open-source project, PVDeg welcomes community contributions through GitHub issues and pull requests that support improvements to the codebase, documentation, and material-property databases.

Statement of Need

As PV deployment expands, especially into new and demanding operational environments, material degradation poses a challenge to the lifetime of PV modules. Modeling degradation is crucial for anticipating performance losses, guiding material selection, and enabling proactive maintenance strategies that extend the operational lifetime of PV modules in diverse environments. Currently, no open-source software combines physics-based degradation mechanism modeling with uncertainty quantification and geospatial scaling. This repository offers a powerful set of tools for PV reliability researchers, materials scientists, and engineers in both industry and research laboratories.

State of the Field

Existing PV modeling tools such as `pvlb-python` (Holmgren, Hansen, & Mikofski, 2018) (Anderson et al., 2023) and SAM (Blair et al., 2018) can simulate system energy yield, but not degradation. Degradation rates and performance loss rates may be extracted from PV performance data using packages such as `RdTools` (Deceglie et al., 2026) and `PVplr` (Curran et

al., 2020). However, these packages adopt a top-down approach whereby temporally-resolved system performance is known and degradation rates are extracted. In this case, the cause of degradation is not always clear. PVDeg addresses this gap with a bottom-up approach, which starts with the known physics of a degradation mechanism and evaluates the associated impact on the performance of a PV system. PVDeg provides modular degradation models, material databases, and uncertainty quantification workflows. PVDeg supports both research and industry use by automating degradation modeling, and enabling reproducible studies of module lifetime and performance worldwide. It also supports ongoing standardization work, including contributions to IEC TS 63126 (International Electrotechnical Commission, 2020) through the geospatial mapping of optimal PV panel standoff distance. Defined as the distance between the module underside and the roof surface, the standoff distance is a critical characteristic of roof-mounted PV systems in particular, and must be optimized for cooling, fire safety, and structural clearance for safe and practical system mounting. The relevance of PVDeg for both research and industry renders it an important component of a growing ecosystem of open-source tools for solar energy, several of which are reviewed in Ref. (Holmgren, Hansen, Stein, et al., 2018).

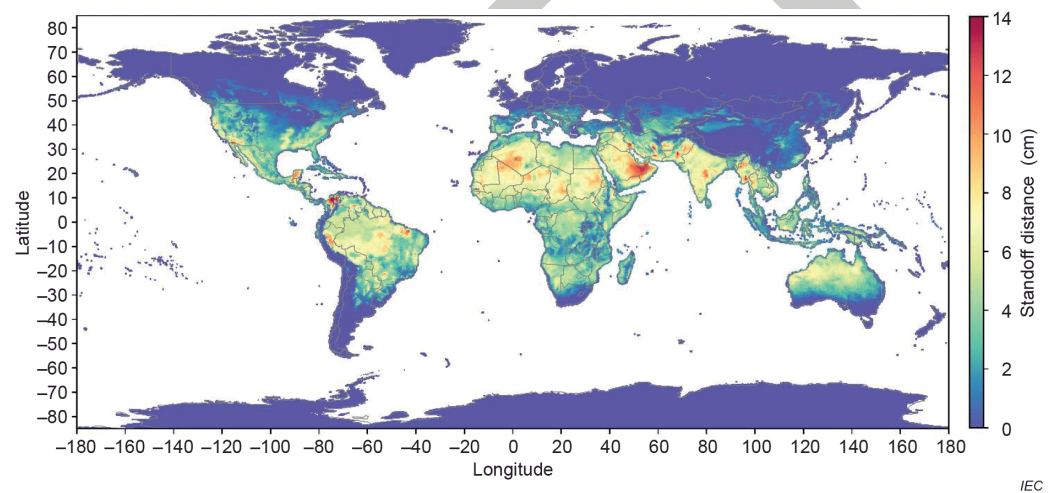


Figure 1: Example of geospatial degradation modeling in PVDeg: calculated standoff distances for IEC TS 63126 across the continental U.S.

Software Design

PVDeg is organized as a layered architecture built around a common data convention. At its foundation is a library of modular *core functions*, each modeling a single environmental stressor (e.g., temperature, irradiance, relative humidity) or degradation mechanism for one location from a standardized weather DataFrame and location metadata. The core functions share this input convention and return either a timeseries or a summary numeric, therefore a user may call one in isolation or chain the output of one (e.g., module temperature) into another (e.g., a moisture-ingress or LeTID model) to assemble a bottom-up lifetime prediction. Three higher-level layers then orchestrate these same functions without re-implementing their physics: the Scenario class sequences them into reproducible, multi-step pipelines; the geospatial layer vectorizes them across latitude-longitude grids; and the Monte Carlo engine propagates parameter uncertainty through them. This reuse is enabled by a lightweight internal decorator (@geospatial_quick_shape) that records each function's output structure, so code written for a single location can be automatically templated and scaled to continental or global analyses without modification. The subsections below describe these layers in turn, together with the tutorials and open datasets that support them.

71 Core Functions

72 The core API provides dedicated functions for calculating physical degradation
73 mechanisms, accessing material properties and environmental stressors. Examples
74 include `pvdeg.humidity.module()` for moisture ingress modeling (Pickett & Coyle, 2013),
75 and `pvdeg.letid.calc_letid_outdoors()` for modeling light and elevated temperature
76 induced degradation (LeTID) (Karas et al., 2022; Repins et al., 2023). These functions rely on
77 standardized environmental stressors such as temperature, irradiance, and humidity, and can
78 be chained to produce lifetime predictions under realistic field conditions.

79 Scenario Class

80 To simplify complex workflows, PVDeg wraps its core functions into a `Scenario` class that
81 defines locations, module configurations, and degradation mechanisms. This enables user-
82 friendly workflows, simplifying the setup and execution of complex multi-parameter degradation
83 studies. This layer provides an intuitive interface for multiple analyses of different degradation
84 modes, climates, and configurations.

85 Geospatial Analysis

86 The geospatial analysis layer enables large-scale spatial analyses by automatically distributing
87 degradation calculations across geographic regions using parallel processing and advanced data
88 structures. It integrates environmental data from NSRDB and PVGIS and automates sampling
89 across latitude-longitude grids to produce maps, such as standoff distance distribution used
90 in IEC TS 63126 compliance studies (International Electrotechnical Commission, 2020). The
91 geospatial layer includes visualization functions for mapping results and supports both uniform
92 and stochastic spatial sampling strategies to balance computational efficiency with geographic
93 coverage. Parallelization routines are compatible with NLR's open-source *GeoGridFusion*
94 framework (Ford et al., 2025; Ford, 2025), allowing users to down-select meteorological
95 datasets efficiently and strategically, and execute computations without high-performance
96 computing access. This capability supports national- and global-scale analyses of degradation
97 phenomena.

98 Monte Carlo Framework

99 Laboratory-to-field extrapolation carries significant uncertainty in kinetic parameters. PVDeg's
100 Monte Carlo engine samples parameter distributions and their correlations to generate
101 thousands of realizations, producing confidence intervals on degradation rates rather than
102 single deterministic values. This capability, described in (Springer et al., 2022), can help
103 quantify uncertainty in complex and non-linear module lifetime predictions, and identify which
104 parameters most strongly affect reliability risk.

105 Tutorials and Tools

106 The tutorials and tools component of PVDeg consists of a comprehensive suite of Jupyter
107 notebooks that demonstrate practical workflows for modeling PV degradation. These notebooks
108 cover core degradation mechanisms, scenario setup, geospatial analysis, and uncertainty
109 quantification, providing step-by-step guidance for both new and advanced users. Each tutorial
110 is designed to be interactive and reproducible, enabling users to explore real-world datasets,
111 customize parameters, and visualize results. The notebooks support comparative studies and
112 integration with external meteorological data sources such as NSRDB and PVGIS.

113 Open datasets

114 A growing component of PVDeg is its compilation of community-driven open datasets for PV
115 degradation modeling. These databases include curated degradation parameters and material

property data, such as kinetic coefficients for common degradation mechanisms, UV-albedo data, and permeation properties for materials (e.g., H₂O, O₂, acetic acid). The datasets are continuously expanded and updated, serving as a growing resource for users to access validated values for modeling and analysis. Users are encouraged to contribute their own data, enhancing the collective knowledge base and supporting reproducible research. The core PVDeg API also provides users with a means to seamlessly query these datasets and use them in their own modeling workflows, analysis, and investigations. The development and maintenance of these degradation databases and associated API calls also support reproducible, reliable, and field-relevant degradation modeling for the PV community.

Research Impact Statement

Since its first release as PV Degradation Tools (Holsapple et al., 2020), PVDeg has been adopted in multiple studies across the PV reliability community:

- Thermal Stability and IEC TS 63126 Compliance: Used to calculate effective standoff distances and generate public maps supporting the IEC TS 63126 standard (International Electrotechnical Commission, 2020).
- Light and Elevated Temperature Induced Degradation (LeTID): Integrated into the international interlaboratory comparison study of LeTID effects in crystalline-silicon modules (Karas et al., 2022) and follow-up analyses of field-aged arrays (Karas, 2024; Repins et al., 2023).
- Geospatial Performance Modeling: Coupled with GeoGridFusion (Ford, 2025) to streamline weather-data storage and spatial queries for large-scale degradation simulations.
- Agrivoltaics and System-Level Modeling: Combined with PySAM (Blair et al., 2018) to assess degradation-driven yield losses and ground-irradiance patterns in dual-use agrivoltaic systems. (Ovatt et al., 2023)
- Material-Property Parameterization: Leveraged in studies of UV-induced polymer degradation (Kempe et al., 2023) and moisture-related failures in encapsulants and backsheets (Coyle, 2011).

These applications highlight PVDeg's versatility as the "PV Library of degradation phenomena" — an open, community-driven platform linking materials science, environmental modeling, and field performance.

Ongoing Development

Version 0.7.4 is the latest stable release, incorporating support for NSRDB PSM v4 weather data and a major restructuring of the Jupyter notebook tutorials for improved usability and clarity.

DuraMAT-funded projects will expand the degradation and material parameter databases using literature searches driven by large language models. In addition, the Scenario class is being developed to support multi-material modules and chained job dependencies, allowing coupled multiphysics modeling of interrelated degradation and transport mechanisms within a single reproducible workflow. This will mitigate the need for users to design and execute multiple individual Scenarios for different degradation pathways and materials.

AI usage disclosure

Generative AI was used to support the creation of docstrings in the software, and changelogs published in release notes. No generative AI tools were used in the writing of this manuscript

or preparation of supporting materials. All outputs created using generative AI were thoroughly vetted by the contributor(s) for accuracy.

Acknowledgements

The authors gratefully acknowledge Derek Holsapple, Matthew Brown, and Aidan Wesley for their foundational contributions to the PVDeg project, as well as the broader open-source community for their continued support and contributions

This work was authored by the National Laboratory of the Rockies for the U.S. Department of Energy (DOE), operated under Contract No. DE-AC36-08GO28308. Funding provided as part of the Durable Module Materials Consortium 2 (DuraMAT 2), funded by the U.S. Department of Energy Office of Critical Minerals and Energy Innovation Integrated Energy Systems Office, agreement number 38259. The views expressed in the article do not necessarily represent the views of the DOE or the U.S. Government. The U.S. Government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

References

- Anderson, K. S., Hansen, C. W., Holmgren, W. F., Jensen, A. R., Mikofski, M. A., & Driesse, A. (2023). Pvlb python: 2023 project update. *Journal of Open Source Software*, 8(92), 5994. <https://doi.org/10.21105/joss.05994>
- Blair, N., DiOrio, N., Freeman, J., Gilman, P., Janzou, S., Neises, T., & Wagner, M. (2018). *System advisor model (SAM) general description (version 2017.9.5)*. National Renewable Energy Laboratory (NREL). <https://doi.org/10.2172/1440404>
- Coyle, D. J. (2011). Life prediction for CIGS solar modules part 1: Modeling moisture ingress and degradation. *Progress in Photovoltaics: Research and Applications*. <https://doi.org/10.1002/pip.1172>
- Curran, A. J., Burleyson, T. L., Rath, K., Xin, A. S., Lindig, S., Moser, D., Stein, J., Bruckman, L. S., & French, R. H. (2020). Pvlpr: R package implementation of multiple filters and algorithms for time-series performance loss rate analysis. *2020 47th IEEE Photovoltaic Specialists Conference (PVSC)*, 2086–2090. <https://doi.org/10.1109/PVSC45281.2020.9300807>
- Deceglie, M. G., Anderson, K. S., Shinn, A., Nag, A., Mikofski, M. A., Springer, M., Yan, J., Perry, K., Villamar, S., Vining, W., Kimball, G., Ruth, D., Moyer, N., Nguyen, Q., Jordan, D., Muller, M., & Deline, C. (2026). *RdTools* (Version v3.1.0). National Renewable Energy Laboratory (NREL). <https://doi.org/10.5281/zenodo.1210316>
- Ford, T. (2025). *GeoGridFusion (open-source geospatial toolkit for solar data integration) [SWR-25-19]*. National Renewable Energy Laboratory (NREL), Golden, CO, United States; National Renewable Energy Laboratory (NREL) GitHub Repository. <https://doi.org/10.11578/dc.20250311.1>
- Ford, T., Ovaite, S., Springer, M., Musleh, Y. J. K., & Kempe, M. (2025). Simplifying geospatial workflows for PV modeling with GeoGridFusion. *2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC)*, 1572–1574. <https://doi.org/10.1109/PVSC59419.2025.11132420>
- Holmgren, W. F., Hansen, C. W., & Mikofski, M. A. (2018). pvlb python: a python package for modeling solar energy systems. *Journal of Open Source Software*, 3(29), 884. <https://doi.org/10.21105/joss.00884>
- Holmgren, W. F., Hansen, C. W., Stein, J. S., & Mikofski, M. A. (2018). Review of open

- 205 source tools for PV modeling. *2018 IEEE 45th Photovoltaic Specialists Conference*.
 206 <https://doi.org/10.5281/zenodo.1401378>
- 207 Holsapple, D., Kempe, M., Ovaith, S., Springer, M., Brown, M., & Ford, T. (2020). *PV*
 208 *degradation tools [SWR-20-71]*. National Renewable Energy Laboratory (NREL), Golden,
 209 CO, United States; National Renewable Energy Laboratory (NREL) GitHub Repository.
 210 <https://doi.org/10.11578/dc.20200714.1>
- 211 International Electrotechnical Commission. (2020). *IEC TS 63126: Guidelines for assessing the*
 212 *lifetime of photovoltaic modules* [Technical Specification]. International Electrotechnical
 213 Commission.
- 214 Karas, J. (2024). Energy yield loss due to LETID. *Proceedings of the 14th SiliconPV Conference*.
 215 <https://www.osti.gov/biblio/2406177>
- 216 Karas, J., Repins, I. L., Berger, K. A., Kubicek, B., Jiang, F., Zhang, D., Jaubert, J.-N.,
 217 Cueli, A. B., Sample, T., Jaeckel, B., Pander, M., Fokuhl, E., Koentopp, M. B., Kersten,
 218 F., Choi, J.-H., Bora, B., Banerjee, C., Wendlandt, S., Erion-Lorico, T., ... Maaroufi,
 219 H. (2022). Results from an international interlaboratory study on light- and elevated
 220 temperature-induced degradation in solar modules. *Progress in Photovoltaics: Research*
 221 *and Applications*. <https://doi.org/10.1002/pip.3573>
- 222 Kempe, M. D., Hacke, P., Morse, J., Owen-Bellini, M., Holsapple, D., Lockman, T., Hoang,
 223 S., Okawa, D., Lance, T., & Ng, H. H. (2023). Highly accelerated UV stress testing
 224 for transparent flexible frontsheets. *IEEE Journal of Photovoltaics*, 13(3), 1823–1823.
 225 <https://doi.org/10.1109/JPHOTOV.2023.3249407>
- 226 Ovaith, S., Kinzer, A., Boyd, M., Jones, J., Deline, C., & Macknick, J. (2023). Validating view-
 227 factor approach and spatial albedo models for bifacial and AgriPV modeling. *Proceedings*
 228 *of the 2023 IEEE 50th Photovoltaic Specialists Conference (PVSC)*. <https://doi.org/10.1109/PVSC48320.2023.10359773>
- 230 Pickett, J. E., & Coyle, D. J. (2013). Hydrolysis kinetics of condensation polymers under
 231 humidity aging conditions. *Polymer Degradation and Stability*, 98(7), 1311–1320. <https://doi.org/10.1016/j.polymdegradstab.2013.04.001>
- 233 Repins, I. L., Jordan, D. C., Woodhouse, M., Theristis, M., Stein, J. S., Seigneur, H. P., Colvin,
 234 D. J., Karas, J. F., McPherson, A. N., & Deline, C. (2023). Long-term impact of light-
 235 and elevated temperature-induced degradation on photovoltaic arrays. *MRS Bulletin*, 48,
 236 589–601. <https://doi.org/10.1557/s43577-022-00438-8>
- 237 Springer, M., Jordan, D. C., & Barnes, T. M. (2022). Future-proofing photovoltaics module
 238 reliability through a unifying predictive modeling framework. *Progress in Photovoltaics:*
 239 *Research and Applications*, 1–8. <https://doi.org/10.1002/pip.3645>